

LEARNING WHICH FUTURE TO IMAGINE: ONLINE REINFORCEMENT LEARNING FOR VIDEO-ACTION WORLD MODELS

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Paper under double-blind review

ABSTRACT

Video-action world models can control robots by imagining a visual future and decoding actions that realize it. Pretraining, however, does not ensure that the imagined future is the one that completes a downstream task. We introduce WAM-RL, an online reinforcement learning method that improves the visual imagination process itself. Starting from a deterministic flow-based video sampler, WAM-RL injects replayable Gaussian noise to obtain stochastic future paths, executes the actions induced by each imagined future with a frozen inverse-dynamics/action module, and assigns the final simulator success verdict back to the sampled video path. Four futures generated from the same robot state form a group-relative policy update. A clipped PPO objective trains only a rank-8 LoRA inside the world model’s video output projection and a 20-step noise schedule; the world-model backbone, base video projection, action head, inverse-dynamics module, and selector remain frozen. On eight LIBERO tasks, the first online round contains 1,280 complete episodes across ten independent seeds, with 289 successes, 991 failures, and no infrastructure exclusions. The resulting 320 same-state groups contain 129 groups with both successful and failed futures, and all ten policy updates produce deployable R1 checkpoints. These results establish an end-to-end route for sparse terminal reward to directly reshape the visual predictions that drive robot behavior.

1 INTRODUCTION

Video world models offer a direct representation of physical consequence: given a scene and an instruction, they can predict what the world should look like after a successful interaction. Recent video-action models turn this representation into a policy by jointly predicting future frames and robot actions (Li et al., 2026). This formulation is attractive because the visual future exposes the policy’s intent. It also creates a distinct failure mode. A pretrained model may produce a plausible future that does not satisfy the task, and the resulting action sequence faithfully executes the wrong imagination.

The standard deterministic sampler compounds this problem. Flow-matching models define an ordinary differential equation (ODE) that maps one noise initialization to one trajectory (Lipman et al., 2023). A single path provides little within-state exploration and no convenient likelihood for asking which alternative future should become more probable. Selecting among a fixed set of predictions does not solve the underlying issue: the generator itself continues to assign probability mass to unhelpful visual futures.

We instead treat the internal video generator as the policy to be improved. Following the likelihood construction of ReinFlow (Zhang et al., 2025), we inject learnable noise into the discrete flow path. The resulting stochastic process produces replayable video trajectories and admits an exact path likelihood. From each robot state, we sample four imagined futures, decode and execute their actions, and use only the final simulator verdict as reward. This produces the comparison needed for sparse-reward credit assignment: successful and unsuccessful futures begin from the same state and differ only in the sampled generative path.

The update is deliberately localized. We insert a rank-8 LoRA (Hu et al., 2021) into the transformer’s internal video output projection and optimize it with a clipped PPO objective (Schulman et al.,

2017). The pretrained world-model backbone, original projection, video decoder, text encoder, action head, inverse-dynamics/action flow, and selector are frozen. Consequently, improvement cannot be attributed to retraining a downstream selector or action policy. Reward changes what the model imagines, and the existing action machinery reacts to that changed visual future.

The experimental protocol targets tasks with non-saturated success rates. Eight LIBERO tasks (Liu et al., 2023) are selected from a stochastic screen, then evaluated over four states and four future paths for each of ten training seeds. Three collect-update rounds are planned. The completed R0 round contains 1,280 simulator-confirmed episodes and yields ten independently trained R1 checkpoints. The ongoing R1 collection tests whether these visual-head updates translate into improved behavior on a fresh state panel.

Our current contribution is an operationally complete visual-RL method rather than an action-level surrogate:

- We adapt online flow-policy optimization to the *internal video generation path* of a video-action world model.
- We construct same-state groups of stochastic imagined futures and assign sparse terminal simulator reward directly to their replayable path likelihoods.
- We isolate the mechanism by training only video-head LoRA parameters and the noise schedule while freezing the action flow and selector.
- We demonstrate full online closure at scale: 1,280 R0 episodes, 320 reward groups, ten independent PPO updates, and ten deployable R1 checkpoints with no infrastructure records in the training denominator.

2 BACKGROUND

Video-action world models. LingBot-VA frames robot control as causal world modeling and jointly learns future visual dynamics and action inference (Li et al., 2026). In the released LIBERO model used here, the transformer hidden representation feeds separate video and action projections. Our intervention is placed in the video projection. The action projection and downstream action flow remain frozen.

From ODE sampling to stochastic policy paths. Flow matching learns a velocity field whose ODE transports a simple source distribution to a data distribution (Lipman et al., 2023). ReinFlow converts deterministic flow-policy sampling into a discrete stochastic process by injecting learnable noise, enabling exploration and path-likelihood policy gradients (Zhang et al., 2025). WAM-RL applies this idea to generated video futures rather than directly to an action-flow policy.

3 METHOD

3.1 TRAINABLE VISUAL POLICY

Let h_k denote the transformer hidden state at flow step k . The frozen video projection has weight W_0 . We insert a low-rank residual

$$v_\phi(h_k) = W_0 h_k + \frac{\alpha}{r} B A h_k, \quad r = \alpha = 8, \quad (1)$$

where A and B are the only trainable video-projection matrices. The original projection weight and bias are unchanged.

3.2 REPLAYABLE STOCHASTIC FUTURE GENERATION

The deterministic Euler discretization of the pretrained video flow is

$$z_{k+1} = z_k + \Delta t f_{\theta, \phi}(z_k, t_k; c), \quad (2)$$

WAM-RL: terminal reward reshapes visual imagination

Train the internal video path; keep action generation and selection fixed.

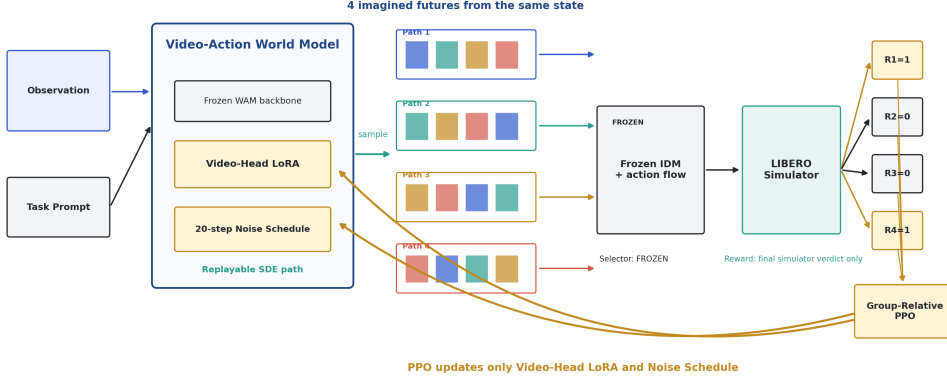


Figure 1: **WAM-RL updates imagination, not selection.** Four replayable video futures are sampled from the same state. A frozen action module executes each future, and the terminal simulator verdict trains only the internal video-head LoRA and stochastic noise schedule.

where c contains the observation and language instruction, θ is frozen, and ϕ denotes the LoRA parameters. We replace it with

$$z_{k+1} = z_k + \Delta t f_{\theta, \phi}(z_k, t_k; c) + \sigma_{\psi, k} \sqrt{\Delta t} \epsilon_k, \quad \epsilon_k \sim \mathcal{N}(0, I), \quad (3)$$

using a learnable 20-step schedule σ_{ψ} . Every noise seed and transition is recorded, so the complete imagined future can be replayed. Under the Gaussian transition, its path log likelihood is

$$\log \pi_{\phi, \psi}(\tau | c) = -\frac{1}{2} \sum_k \left[\frac{\|z_{k+1} - z_k - \Delta t f_{\theta, \phi}(z_k, t_k; c)\|_2^2}{\sigma_{\psi, k}^2 \Delta t} + d \log(2\pi \sigma_{\psi, k}^2 \Delta t) \right]. \quad (4)$$

3.3 SIMULATOR-GROUNDED REWARD

The frozen inverse-dynamics/action flow converts the imagined visual state into a robot action chunk. The LIBERO simulator executes these actions for up to 800 environment steps. Reward is assigned only after execution:

$$R(\tau) = \begin{cases} 1, & \text{the final simulator verdict is success,} \\ 0, & \text{otherwise.} \end{cases} \quad (5)$$

No hidden task predicate, success label, or future simulator information is visible to the planner, generator, action module, or prompt.

3.4 SAME-STATE GROUP-RELATIVE PPO

For each task-state pair, we sample $K = 4$ stochastic futures. Their group-relative advantages are

$$A_i = R_i - \frac{1}{K} \sum_{j=1}^K R_j. \quad (6)$$

The likelihood ratio for the recorded path is

$$\rho_i(\phi, \psi) = \exp(\log \pi_{\phi, \psi}(\tau_i | c) - \log \pi_{\text{old}}(\tau_i | c)). \quad (7)$$

We optimize four epochs of the clipped objective

$$\mathcal{L}_{\text{PPO}} = -\mathbb{E}_i [\min(\rho_i A_i, \text{clip}(\rho_i, 1 - \varepsilon, 1 + \varepsilon) A_i)] + \lambda \mathcal{L}_{\text{anchor}}, \quad \varepsilon = 0.1. \quad (8)$$

Groups with identical reward have zero centered advantage. Groups containing both success and failure directly increase the relative likelihood of the successful imagined future.

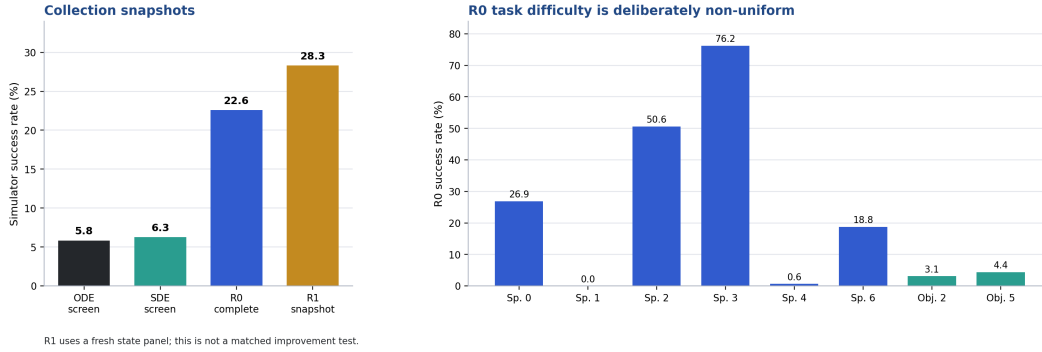


Figure 2: **Completed and live evidence.** Stochastic sampling creates reward-bearing paths, while R0 spans a deliberately broad range of task difficulty. R1 is an incomplete fresh-state snapshot, not a matched before/after comparison.

4 EXPERIMENTAL SETUP

Benchmark and tasks. Experiments use the released 800-step LIBERO simulator horizon. A preliminary screen covers spatial, object, and goal suites. The online set contains eight tasks: six spatial bowl-placement tasks and two object-to-basket tasks. Five tasks have observed success/failure splits in screening; three additional tasks provide same-family exploration headroom. Two tasks are held out for later evaluation.

Online protocol. Each training seed and online round uses

$$8 \text{ tasks} \times 4 \text{ states} \times 4 \text{ future paths} = 128 \text{ episodes.}$$

Ten independent seeds produce 1,280 episodes per round. State panels are disjoint across R0, R1, R2, and final evaluation. We run three collect-update rounds. The current implementation uses 20 resumable rollout workers across eight GPUs.

Optimization. The video-head LoRA learning rate is 10^{-4} ; the noise-schedule learning rate is 10^{-3} . The stochastic standard deviation is initialized to 0.01 and constrained to $[0.002, 0.03]$. We record four representative 20-transition flow chains per episode. The world-model backbone, original video projection, VAE decoder, text encoder, action head, inverse-dynamics flow, and selector remain frozen.

Evaluation. The main behavioral endpoint is simulator success rate. Final comparisons use matched fresh states for the base model and R1/R2/R3 checkpoints, reported per task and across ten seeds. Supporting diagnostics include same-state success/failure group count, successful-over-failed path ranking, replay error, approximate KL, LoRA update norm, learned noise schedule, tokens, policy calls, and wall time.

5 RESULTS

5.1 STOCHASTIC VISUAL PATHS EXPOSE REWARD-BEARING ALTERNATIVES

The main R0 collection is complete with no infrastructure records. Success is highly task dependent: the strongest spatial tasks reach 122/160 and 81/160 successes, while the two object tasks reach 7/160 and 5/160. This spread is useful for online learning because the data contain both reward-bearing paths and difficult failure modes rather than only saturated tasks.

Across ten seeds, R0 contains 320 same-state groups. Of these, 129 contain both successful and failed futures. These groups provide direct visual credit: the observation, task, and simulator state are held fixed, while the sampled video path and induced behavior differ.

Table 1: Completed and live simulator collections. R1 is an August 26, 2026, 07:00 UTC snapshot and uses a new state panel, so its raw rate is not a matched before/after comparison.

Collection	Episodes	Success	Failure	Success rate
Deterministic ODE screen	120	7	113	5.8%
Stochastic SDE screen	239	15	224	6.3%
R0 online collection	1,280	289	991	22.6%
R1 online snapshot	212	60	152	28.3%

Table 2: Aggregate diagnostics from ten completed R0 → R1 updates.

Diagnostic	Value
Independent trained checkpoints	10 / 10
Variable-reward groups	129 / 320
Mean LoRA update norm	0.03027
Maximum replay log-probability error	0.01619
Maximum approximate KL	3.42×10^{-9}
Mean log-probability change, success paths	1.149×10^{-4}
Mean log-probability change, failed paths	1.021×10^{-4}
Final mean noise standard deviation	0.0100229

5.2 TERMINAL REWARD UPDATES THE INTERNAL VIDEO HEAD

All ten optimizers complete four epochs without NaNs or clipping events and produce deployable internal-LoRA checkpoints. Successful paths receive a larger average log-probability increase than failed paths. The learned noise schedule remains close to its initialization, while the video-head LoRA has a consistent nonzero update norm. At this stage, the principal learned change is therefore in the visual output mapping, while stochastic noise supplies the exploration and likelihood construction.

5.3 ONLINE CLOSURE

The ten R1 checkpoints are loaded back into the WAM server and are currently collecting the next 1,280-episode state panel. This closes the intended loop: stochastic visual rollout, simulator reward, path-likelihood PPO, internal video-head update, checkpoint deployment, and fresh online interaction. The paper’s final behavioral claim will be based on matched evaluation states rather than the unmatched R0 and R1 collection rates in Table 1.

6 DISCUSSION

Why update the video head? The imagined video is the causal interface between task language and action. If the action module is frozen, changing the future prediction changes the visual evidence from which continuation actions are inferred. This targets the upstream source of behavior while preserving the pretrained action prior.

Why not train a selector? A selector can choose among available candidates but cannot improve the candidate distribution. WAM-RL instead increases the probability of rewarded futures inside the generator. The selector is frozen to keep this mechanism identifiable.

What the current evidence establishes. The completed evidence establishes stochastic exploration, same-state success/failure forks, exact replay, sparse terminal credit assignment, nonzero internal video-head updates, and successful redeployment of ten trained checkpoints. Matched R0/R1/R2/R3 evaluation is the remaining test of behavioral improvement.

7 LIMITATIONS

The current online task set is dominated by spatial bowl placement because object and goal tasks are substantially harder under the released policy. Raw collection rates across rounds use disjoint state panels and must not be interpreted as matched policy improvement. The learned noise schedule changes only slightly in R1, so stronger claims about adaptive exploration require later rounds or a dedicated fixed-noise control. Finally, the terminal binary reward is intentionally simple; denser stage rewards may improve credit assignment but would introduce additional reward-design assumptions.

8 CONCLUSION

WAM-RL turns a video-action world model into an online visual policy. It samples replayable stochastic future videos, executes their induced actions, and uses the terminal simulator verdict to update only the internal video generation head. The completed first round provides 1,280 simulator episodes, 129 informative same-state reward groups, and ten deployed R1 checkpoints. This establishes a direct route from sparse task success to the future that the world model learns to imagine.

DATA AVAILABILITY

The release package will include protocol files, exact task/state/path manifests, per-episode simulator verdicts, videos, traces, replayable flow chains, optimizer analyses, and trained LoRA checkpoints. Interrupted and infrastructure records are retained but excluded from behavioral denominators.

ETHICS AND BROADER IMPACT

The study uses simulated robot manipulation and does not involve human subjects or personal data. Improving visual world-model adaptation may reduce the amount of physical-robot data required for downstream tasks, but deployment on real robots still requires independent safety constraints and validation.

AUTHOR CONTRIBUTIONS

Anonymous for double-blind review.

CONFLICT OF INTEREST AND FUNDING

The authors declare no conflict of interest. Funding information is withheld during anonymous review.

AI USE STATEMENT

AI tools assisted with manuscript drafting and figure generation. The authors are responsible for the method, implementation, experiments, citations, and claims.

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A TASK-LEVEL R0 RESULTS

Table 3: Completed R0 simulator verdicts.

LIBERO task	Success	Episodes	Rate
Spatial 0	43	160	26.9%
Spatial 1	0	160	0.0%
Spatial 2	81	160	50.6%
Spatial 3	122	160	76.3%
Spatial 4	1	160	0.6%
Spatial 6	30	160	18.8%
Object 2	5	160	3.1%
Object 5	7	160	4.4%

B PROMPT

Each task uses its public LIBERO instruction followed by:

Predict a physically consistent successful future: move the gripper deliberately, establish stable grasp or contact with the named object, complete the stated spatial or articulated relation fully, avoid disturbing unrelated objects, and finish in a stable completed state.

The prompt contains no hidden success predicate or simulator feedback.